



Automatic mandible segmentation from CT image using 3D fully convolutional neural network based on DenseASPP and attention gates

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Abstract

Purpose In cranio-maxillofacial surgery, it is of great clinical significance to segment mandible accurately and automatically from CT images. However, the connected region and blurred boundary in teeth and condyles make the process challenging. At present, the mandible is commonly segmented by experienced doctors using manually or semi-automatic methods, which is time-consuming and has poor segmentation consistency. In addition, existing automatic segmentation methods still have problems such as region misjudgment, low accuracy, and time-consuming.

Methods For these issues, an automatic mandibular segmentation method using 3d fully convolutional neural network based on densely connected atrous spatial pyramid pooling (DenseASPP) and attention gates (AG) was proposed in this paper. Firstly, the DenseASPP module was added to the network for extracting dense features at multiple scales. Thereafter, the AG module was applied in each skip connection to diminish irrelevant background information and make the network focus on segmentation regions. Finally, a loss function combining dice coefficient and focal loss was used to solve the imbalance among sample categories.

Results Test results showed that the proposed network obtained a relatively good segmentation result, with a Dice score of $97.588 \pm 0.425\%$, Intersection over Union of $95.293 \pm 0.812\%$, sensitivity of $96.252 \pm 1.106\%$, average surface distance of 0.065 ± 0.020 mm and 95% Hausdorff distance of 0.491 ± 0.021 mm in segmentation accuracy. The comparison with other segmentation networks showed that our network not only had a relatively high segmentation accuracy but also effectively reduced the network's misjudgment. Meantime, the surface distance error also showed that our segmentation results were relatively close to the ground truth.

Conclusion The proposed network has better segmentation performance and realizes accurate and automatic segmentation of the mandible. Furthermore, its segmentation time is 50.43 s for one CT scan, which greatly improves the doctor's work efficiency. It will have practical significance in cranio-maxillofacial surgery in the future.

Keywords Mandible segmentation · Fully convolutional neural network · DenseASPP · Attention gates

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Introduction

The mandible is the only bone with motor function in the maxillofacial region. Its normal function is critical to ensure the actions of chewing, swallowing, pronouncing and even to maintain the shape of the face. Therefore, it is necessary to accurately segment the mandible in many clinical operations, such as mandibular reconstruction surgery [1], oral implantation [2], maxillofacial surgery planning [3], head and neck tumor radiotherapy [4], etc. At present, doctors usually use manual or semi-automatic methods to segment the mandible before the surgery. For manual segmentation, it requires experienced doctors to depict contours in each slice of the computed tomography (CT) data. Semi-automatic segmentation is an interactive segmentation method based on threshold segmentation, region growth, level set, etc. The doctors have to manually set the initial position, adjust the parameters constantly, and modify the segmentation results [5, 6]. These segmentation methods are time-consuming, less efficient, and have poor segmentation consistency.

Therefore, a method that can accurately and automatically segment the mandible is desired. However, when patients take their CT scans, they commonly need to fit together the upper and lower teeth. The connected region and the blurred boundary between the upper and lower teeth all increase the difficulty of segmentation. Besides, the regions of upper and lower jaw bones are also overlapped at the position of the condyle, which also makes it difficult to be segmented. At the same time, the morphological difference of the mandible from different patients also limits the robustness of the segmentation method. The above factors are the challenges of automatic segmentation of the mandible.

For the segmentation of the mandible, some scholars have proposed some automatic methods based on traditional segmentation methods. For example, Gass et al. [7] proposed an automatic segmentation method of the mandible combining image segmentation based on Markov random field with registration using multiresolution non-rigid atlas, but its speed was relatively slow. For mandibular CBCT images, an automatic segmentation method of unsupervised spatially consistent clustering forest was proposed by Pei et al. [8], which was better than supervised and unsupervised forest methods. Fan et al. [9] applied Marker-based watershed transformation to automatically segment CBCT images of the mandible. Taghizadeh et al. [10] established a statistical shape model of the mandible and used local template matching to realize the automatic segmentation of the mandible. Although the above

traditional segmentation methods have achieved a certain segmentation effect, their speeds are relatively slow, and a larger number of parameters are to be set. Moreover, when the morphological structure of the mandible is different or the boundary is blurred, these methods are easy to fail.

With the rapid development of deep learning, more and more scholars have applied the technique to segment medical images [11]. For the automatic segmentation of the mandible, some researches have been carried out. For the segmentation of mandible in organs-at-risks of head and neck, Ibragimov et al. [12] used a convolutional neural network (CNN) to classify CT image blocks and then smoothed the result using the Markov random field algorithm. This method realized automatic segmentation of mandibles, but the effect was not good on the contours. Yan et al. [13] proposed an automatic symmetrical convolutional neural network to segment mandible using the end-to-end symmetry of convolution and deconvolution. Qiu et al. [14] proposed an automatic segmentation method of mandible based on 2.5D U-Net. It first automatically segmented the 2D slices of the mandible from three directions and then combined the results to realize the automatic procedure. However, this method increased the network segmentation time. Gou et al. [15] added the channel and spatial attention module to the network to improve the segmentation performance and realized the automatic segmentation of multiple organs in the head and neck. Above deep learning methods have achieved automatic segmentation of the mandible, but there still are some problems such as misjudgment at the common regions of the upper and lower teeth and the condyle, and the robustness of the segmentation needed to be improved.

To automatically perform mandible segmentation from CT images with greater accuracy, this paper proposed an automatic segmentation method using 3d fully convolutional neural network (FCN) based on densely connected atrous spatial pyramid pooling (DenseASPP) [16] and attention gates (AG) [17]. Firstly, we established a fully convolutional neural network based on an end-to-end symmetry network. And, the DenseASPP module was added to the network to achieve multi-scale feature search and to improve network segmentation performance. Secondly, we included the AG module in the network, which optimized the features with the performance of the attention gates. More importantly, we used the loss function combining focal loss [18] and dice coefficient loss, where the focal loss was used to optimize the segmentation network by solving the imbalance of sample categories. Finally, the proposed segmentation network not only achieved effective automatic segmentation of the mandible, but also had a better segmentation accuracy than other segmentation networks.

Methodology

The structure of the segmentation network

For the reason of blur boundary of the teeth and condyle, and various shapes of the mandible, traditional segmentation methods cannot achieve an effective automatic segmentation. In this paper, we apply the FCN method for mandible segmentation due to its performance in the automatic segmentation of medical images. There are many structures of deep learning segmentation networks such as 2D, 2.5D, 3D, and their combination. Though the 2D and 2.5D networks can segment 2D CT slices using small GPUs, the correlation information between CT slices isn't considered in the network. To make full use of the correlation information for automatic segmentation of the mandible, the structure of 3D FCN is chosen in this work.

The structure of the proposed network is shown in Fig. 1, which is based on an FCN with an asymmetrical structure of encoding and decoding processes and contains 9 convolutional layers. In each convolutional layer, all filters are 3D convolutions with convolution kernels of $3 \times 3 \times 3$, in which the ReLU activation function and feature normalization operations are included. In the encoding process, each convolutional layer is composed of stacked convolution and down-sampling convolution. Among them, two 3D convolution filters are stacked to compose

the stacked convolution for expanding their receptive field sizes of search features. The stride of 1 can keep the size and channel of the search feature unchanged. In down sampling, the convolution is used to replace the maximum pooling for better extracting features. The strides are set as 2. Therefore, the size of the feature map will be decreased by half and the channel will be expanded twice during the search process. In the decoding process, each convolutional layer is composed of stacked convolutions and deconvolutions. The stacked convolutions are similar to those in encoding, except that the first convolution needs to decrease the channel of the feature map by half from the previous concatenation. Deconvolution is used to decode and restore feature maps to images, which doubles the feature map and reduces the channel by half. In the entire network, the size of the input patch is $16 \times 256 \times 256$. The feature map will change from $16 \times 256 \times 256$ to $1 \times 16 \times 16$ and the channel will change from 1 to 256 in the encoding process. The decoding is the opposite process.

The main advantages of the proposed network structure include: Firstly, residual connection is applied to each convolutional layer except the middle ones. In those convolutional layers, the input feature maps of the first convolution are added into the output feature maps of the second convolution, which can prevent the loss of features of the same convolutional layer and accelerate the convergence of the network. Then, considering that batch normalization has a poor performance for small batch size, the group normalization (GN) [19] is adopted into

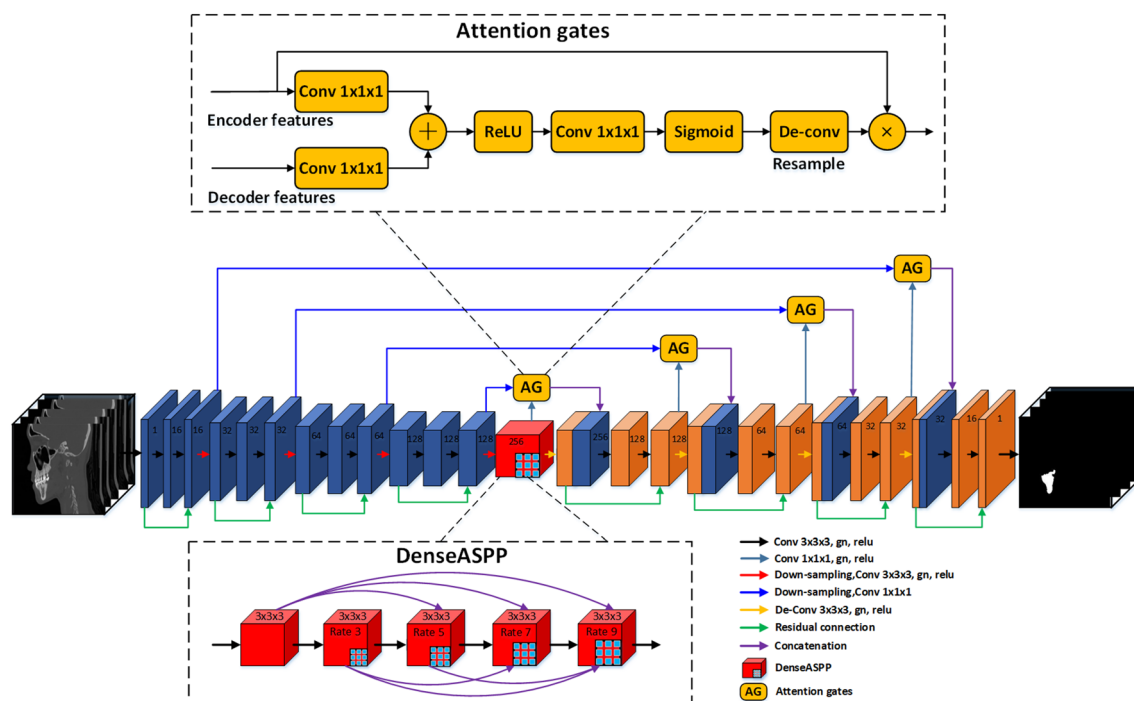


Fig. 1 Proposed network based on dense and ASPP AG

the network for normalizing features more effectively, which also prevents the network from overfitting. Furthermore, the DenseASPP module was applied to the middle convolutional layers for extracting more dense features at multiple scales. Last but not least, we use the AG module to optimize features in each skip connection. This module helps pay more attention to the feature regions that require segmentation while leaving behind irrelevant regions. At the same time, skip connections can reduce the loss of shallow features when the number of network convolutional layers increases.

DenseASPP for extracting features

As the number of network layers increases in the network encoding process, the size of the feature map continues to decrease, and the deep feature information will be difficult to search, which is not conducive to restore the feature map during the decoding process. Therefore, this paper uses the DenseASPP module proposed by Yang et al. [16] to improve feature search in the last layer of the encoding process. It can obtain the multi-scale dense feature maps by concatenating the results from atrous convolutions [20] with different atrous rates. Atrous convolution is that the middle of the convolution kernel is inserted into some 0 value in ordinary convolution, which can make the convolution obtain a larger receptive field size without increasing the convolution kernel parameters. The size of the search field R will be obtained by the following Eq. (1):

$$R = K + (K - 1)(d - 1) \quad (1)$$

where K is the kernel size of atrous convolution and d is the dilation rate of atrous convolution.

By the larger receptive field size, the atrous convolution can search for larger feature maps. When the feature maps from the atrous convolutions with different dilation rates are combined, multi-scale context feature information will be obtained. At the same time, the DenseASPP module also uses the concatenation to merge search feature maps in the pyramid structure to extract dense multi-scale feature maps. The DenseASPP structure is shown in Fig. 1, it consists of 5 atrous convolutions with the atrous rate of 1, 3, 5, 7, and 9, where the convolution kernels are all $3 \times 3 \times 3$. It can achieve dense pyramid connection using these convolutions to extract feature maps sequentially and skip concatenations to combine them, in which the adjacent skips are removed to reduce the amount of calculation. Finally, it can obtain the sums of multi-scale feature map from the receptive field of $3 \times 3 \times 3$, $7 \times 7 \times 7$, $11 \times 11 \times 11$, $15 \times 15 \times 15$, and $19 \times 19 \times 19$.

Attention gates for improving upsampling

Since the mandible is only a part of the CT image, we need to search for the mandible in the entire image during segmentation. However, some irrelevant background information will easily lead to misjudgment. For this reason, it is a practical alternative to add some attention module into the CNN network to make it focus on the segmented regions and minimize distractions from unrelated regions. An effective AG module proposed by Oktay et al. [17], which has achieved a relatively good result in U-Net, is applied to our network to improve the segmentation performance, as shown in Fig. 1. The input features of AG include the encoder features and decoder features. The encoder features will be operated by down sampling, which use a convolution with the convolution kernel of $1 \times 1 \times 1$ and the stride of 2 to reduce the feature size by half and remain the channel unchanged. The decoder features will be operated by a convolution with the convolution kernel of $1 \times 1 \times 1$ to make the channel the same as encoder features. And, the feature vector will be obtained by adding two results in the corresponding elements, which increases the feature weight of the useful region. Subsequently, the feature vector will be compressed by the ReLU activation function and the convolution with $1 \times 1 \times 1$ convolution kernel. Then, for obtaining the weight coefficient of the attention parameter, the feature vector will be normalized to the range $[0, 1]$ by a sigmoid operation. When the weight coefficient is closer to 1, the features are more relevant. Afterward, the feature vector will be resampled to the size of the input encoder feature map. Finally, the final feature map highlighting the segmentation region more will be obtained by multiplying the former result and the feature map from the decoding process. The AG modules are added into each skip concatenation, which makes the segmentation network focus better on the mandible regions in CT images than the background region.

Loss function

Another advantage of our network is the improvement of the loss function. Currently, the common loss functions used in segmentation networks are cross-entropy loss and dice coefficient loss. The loss function of the dice coefficient is more suitable for medical image segmentation, and it can be obtained by Eq. (2). Although these two loss functions can achieve good segmentation results, they are not suitable when the classes of the training data set are not balanced. In order to effectively solve the problem of class imbalance, Lin et al. [18] proposed the loss function of focal loss from Eq. (3). This loss function is improved based on the cross-entropy, and the coefficient will be adjusted by controlling the modulation factor $(1 - p_t)^\gamma$ and the weighting factor α_t . When there are more segmentation errors, the prediction

probability p_i will be smaller, and the modulation factor will be close to 1, which has a greater impact on the loss. When the segmentation error is not much, the prediction probability p_i will be larger, and the modulation factor will be close to 0, which will have less impact on the loss. At the same time, the weighting factor α_i and γ are also added to further adjust the modulation factor. To better segment the mandible, the respective advantages of dice coefficient loss and focus loss are combined. We use the weight parameters a and b to combine two loss functions, and the final loss function can be obtained by Eq. (4). According to the effect of segmentation, the parameters setting is that $\alpha_i = 0.25$, $\gamma = 2$ in the network.

$$\text{Loss}_{\text{Dice}} = 1 - \frac{2 \sum_i^N p_i g_i}{\sum_i^N p_i^2 + \sum_i^N g_i^2} \quad (2)$$

where N is the total number of voxels in input batch, and $i = 1, 2, \dots, N$. p_i is the predicted binary segmentation voxel, and g_i is the ground truth binary voxel.

$$\text{Loss}_{\text{Focal}} = -\alpha_i (1 - p_i)^\gamma \log(p_i) \quad (3)$$

where $\alpha_i \in [0, 1]$ is the weighting factor, $(1 - p_i)^\gamma$ is the modulating factor, $\gamma \geq 0$, and p_i is the probability that the predicted binary segmentation volume belongs to the ground truth binary volume.

$$\text{Loss} = a\text{Loss}_{\text{Dice}} + b\text{Loss}_{\text{Focal}} \quad (4)$$

Experiments and results

Data set and settings

The data set was provided by Shanghai Ninth People's Hospital Affiliated to Shanghai Jiao Tong University School of Medicine. There were a total of 60 CT scans, including 35 CT scans for training data and 25 CT scans for testing data. The sizes of all CT scans were from $512 \times 512 \times 146$ to $512 \times 512 \times 788$ and their spacings were from $0.421 \times 0.421 \times 0.329$ mm to $0.519 \times 0.519 \times 1.0$ mm. The ground truth was produced by experienced doctors. According to the clinical segmentation requirements, they manually delineated the contours of the mandible and erased error regions on each slice of the CT scan using ITK-SANP 3.8 software, and corrected them carefully.

The size of all CT data was normalized to $512 \times 256 \times 256$ for training and testing segmentation network more conveniently. We also removed some CT slices that didn't include the mandible at the sagittal plane direction, but not at the other directions. Considering that putting the whole CT scan into the 3D segmentation network in training, larger GPU

memory and calculations would be required. And, the sagittal plane could show the shape of the mandible better, which was conducive to segmentation. Therefore, the training patch was set to $16 \times 256 \times 256$, which was cut from CT scan along the coronal axis with a stride of 12. The size of testing patch was also $16 \times 256 \times 256$. However, for the convenience of final stitching, it was cut from CT scan along the coronal axis with a stride of 16.

In the network training, the batch size was 1, the learning rate was 0.001, the drop out was 0.5, the optimization algorithm was Adam and the number of training iterations was 15,000. Our code was implemented using TensorFlow2.0, and the experiments were performed on a GPU NVIDIA RTX8000.

Evaluation metrics

To better evaluate the segmentation results, the dice coefficient (Dice), Intersection over Union (IOU), sensitivity, average surface distance (ASD) and 95% Hausdorff distance (HD) were selected as the evaluation metrics of segmentation accuracy, as shown in Eqs. (5)–(8) and (10):

$$\text{Dice} = \frac{2\text{TP}}{\text{FP} + 2\text{TP} + \text{FN}} \quad (5)$$

$$\text{IOU} = \frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}} \quad (6)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (7)$$

where TP are true-positive numbers, FP is false-positive numbers and FN is the false-negative numbers in segmentation results.

$$\text{ASD} = \frac{1}{2} \left(\frac{\sum_{x \in X} \min_{y \in Y} d(x, y)}{|X|} + \frac{\sum_{y \in Y} \min_{x \in X} d(y, x)}{|Y|} \right) \quad (8)$$

$$\vec{d}_{H,r}(X, Y) = K_r \left(\min_{y \in Y} d(x, y) \right), \quad \forall x \in X \quad (9)$$

$$\text{HD} = \max \left(\vec{d}_{H,r}(X, Y), \vec{d}_{H,r}(Y, X) \right) \quad (10)$$

, where X and Y are the ground truth and test regions. K_r denoted the r -th percentile, here we chose 95%. $d(x, y)$ denoted the Euclidean distance of the points of X to the points of Y . $\vec{d}_{H,r}(X, Y)$ denoted the minimum Euclidean distance of the points of X to the points of Y .

Comparison methods

Some segmentation networks were selected for comparison to better evaluate the segmentation effect of the proposed network. As shown in Table 1, the FCN [21], U-Net [22], and V-Net [23] were selected as the comparison method, which are the most commonly used in medical image segmentation. Considering that the proposed network was based on the structure of V-Net, in order to furtherly evaluate the improvement effect of each additional module, we compared our method1, method2, and method3. Among them, method1 added the DenseASPP module on the basis of V-Net, method2 added the focal loss on the basis of method1, and method3 added the AG module on the basis of method2. Method3 was the finally proposed segmentation network. Above networks were all 3D network structures, and their parameters were set same as the proposed network.

The results of mandible segmentation

In the proposed network, the weight parameters a and b affected the segmentation effect. For obtaining the best segmentation effect, we conducted 10 comparison experiments by setting different a and b values, as shown in Fig. 2. The accuracy comparison of the Dice, IOU, and sensitivity showed that the best segmentation accuracy can be obtained when a was 0.7 and b was 0.3. Therefore, the parameters setting of loss function was that $a = 0.7$ and $b = 0.3$.

The final test results showed that the proposed network had a relatively fast segmentation rate, with an average segmentation time of 50.43 s for one CT scan. At the same time, it had a relatively good segmentation accuracy, with a Dice score of $97.588 \pm 0.425\%$, IOU of $95.293 \pm 0.812\%$, sensitivity of $96.252 \pm 1.106\%$, ASD of 0.065 ± 0.020 mm, and 95% HD of 0.491 ± 0.021 mm. As shown in Table 1, the comparison with other networks showed that the proposed network had a significant improvement in the segmentation accuracy of Dice, IOU, sensitivity, ASD, and 95% HD. It had also a relatively low standard deviation, indicating that the fluctuation of segmentation accuracy in

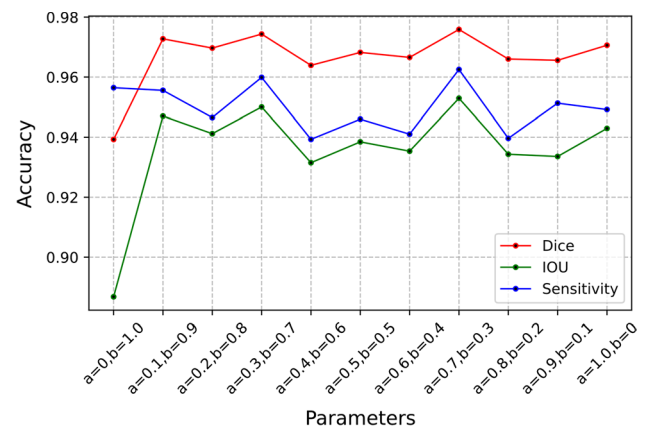


Fig. 2 Line chart of accuracy comparison from the different settings of weight parameters a and b in the loss function

each CT image was less. Figure 3. was their boxplots about the Dice, IOU, sensitivity, ASD, and 95% HD. Among them, in terms of the median of Dice, IOU, and sensitivity, the values were improved by 0.02, 0.037, and 0.04, respectively, using our proposed method compared with V-Net, which is the most efficient traditional network. As for the median of ASD and 95% HD, the values were reduced by 0.146 mm and 0.278 mm, respectively, compared with V-Net. It indicated that our proposed network was significantly better than other networks. Furthermore, the total normal distribution of the proposed network was more concentrated than other methods. The above results showed that our proposed network had significant improvement in segmentation accuracy compared with the other networks.

Figure 4 was a comparison of the contours from ground truth and segmentation results about 4 networks. During the comparison, our proposed network had a better segmentation performance, especially in effectively reducing the misjudgment regions in the condyle and between the teeth. In order to further analyze the mandible segmentation in space, we used Marching Cubes in VTK7.1 to generate 3D reconstructed models from the ground truth and the segmentation results, and calculated their surface distance error. As shown in Fig. 5, the most surface distance

Table 1 Evaluation results from comparison with other methods

Methods	Dice (%)↑	IOU (%)↑	Sensitivity (%)↑	ASD (mm)↓	95% HD (mm)↓
FCN [21]	92.544 ± 0.890	86.135 ± 1.520	90.582 ± 1.965	0.345 ± 0.109	1.120 ± 0.261
U-Net [22]	94.952 ± 2.073	90.460 ± 3.557	90.929 ± 3.459	0.256 ± 0.303	1.128 ± 1.271
V-Net [23]	95.406 ± 0.909	91.230 ± 1.663	92.333 ± 1.436	0.377 ± 0.401	1.768 ± 2.671
Our method1 (DenseASPP)	97.045 ± 0.655	94.268 ± 1.234	94.830 ± 1.417	0.086 ± 0.031	0.561 ± 0.153
Our method2 (DenseASPP + focal loss)	97.061 ± 0.474	94.294 ± 0.899	94.917 ± 1.205	0.092 ± 0.049	0.541 ± 0.129
Our method3 (DenseASPP + AG + focal loss)	97.588 ± 0.425	95.293 ± 0.812	96.252 ± 1.106	0.065 ± 0.020	0.491 ± 0.021

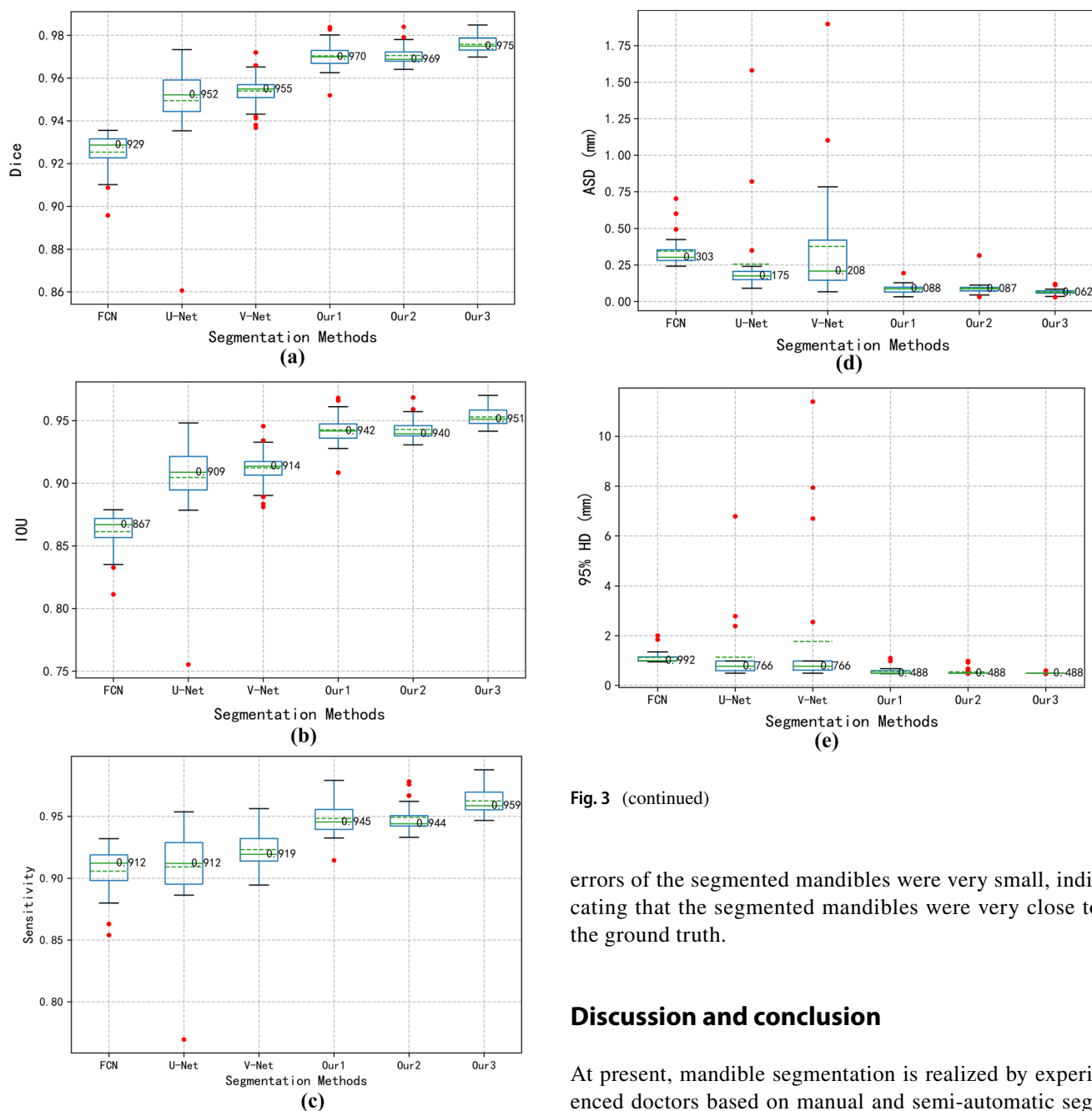


Fig. 3 (continued)

errors of the segmented mandibles were very small, indicating that the segmented mandibles were very close to the ground truth.

Discussion and conclusion

At present, mandible segmentation is realized by experienced doctors based on manual and semi-automatic segmentation methods. However, these methods often require parameters tuning, selection of initial points, and erasing of misjudgment regions, which either results in a long time, less accurate, or complicated operation. Furthermore, the automatic segmentation methods proposed by other scholars still have shortcomings in the accuracy and performance of segmentation.

Therefore, the automatic segmentation method for the mandible in CT image using 3D fully convolutional neural network based on DenseASPP and AG was proposed in this paper. In the network, we used the DenseASPP module to extract dense features at multiple scales and used the AG module to leave behind irrelevant

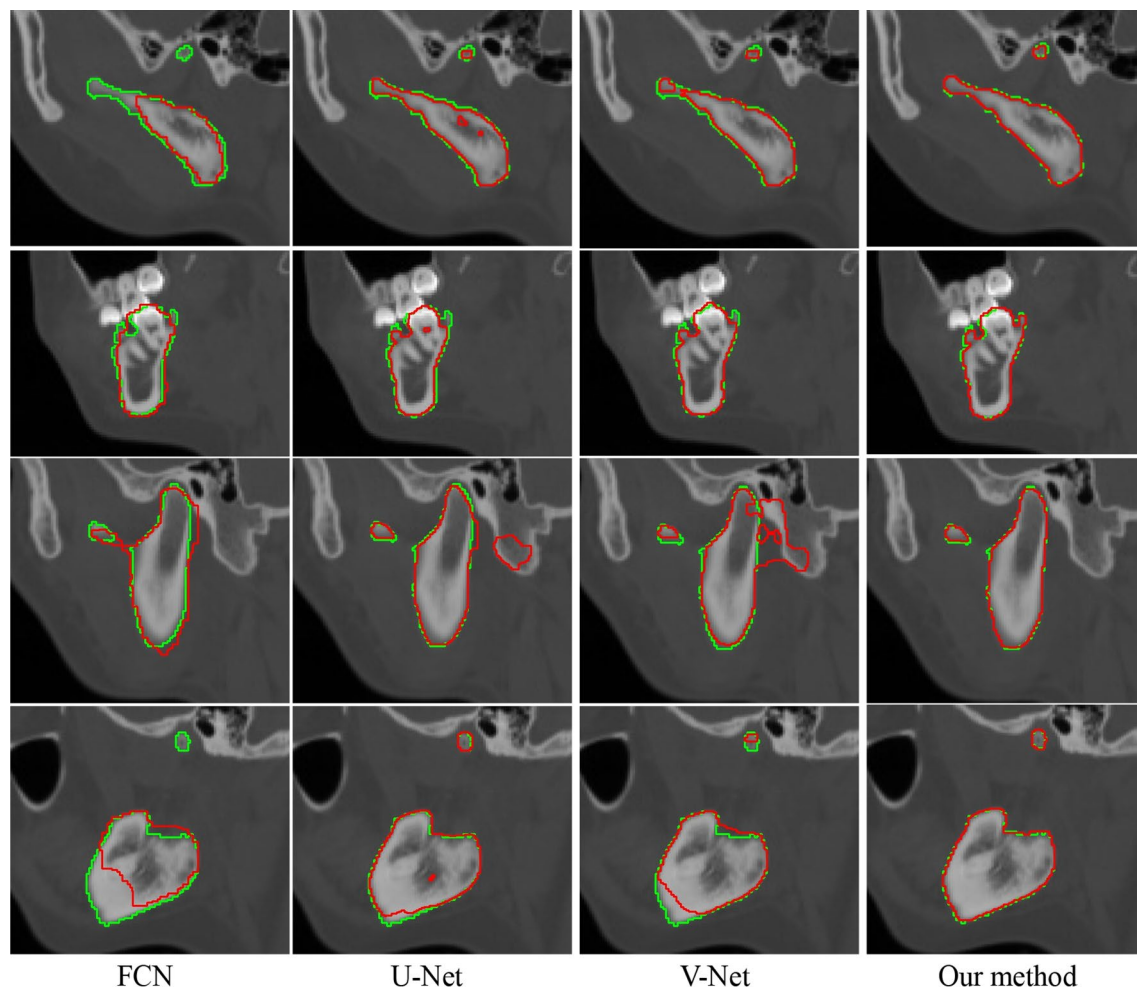


Fig. 4 presents a comparison of the stated segmentation methods with ground truth on 4 selected images. For easy observation, the images are cropped from the complete mandible images. Green lines:

the edges of ground truth image; Red lines: the edges of segmentation results by these methods

background regions, allowing the network to focus more on the segmented regions. Meantime, we used a loss function combining dice coefficient and focal loss to further solve the problem of data imbalance. After training on 35 training sets, the results in 25 test sets showed that the Dice reached $97.588 \pm 0.425\%$, the IOU reached $95.293 \pm 0.812\%$, the sensitivity reached $96.252 \pm 1.106\%$, the ASD reached 0.065 ± 0.020 mm, and the 95% HD reached 0.491 ± 0.021 mm in the segmentation accuracy of our proposed method. Compared with other networks in Table 1 and Fig. 4, the segmentation accuracy was significantly improved, and reduced the misjudgment of the network and improved the segmentation performance in blurred regions of the condyle and between the teeth.

For the segmentation time, the average segmentation time of each CT scan was 50.43 s in the proposed network, which was much faster than manual segmentation of more than 30 min usually. Compared with the 1 min of semi-automatic

segmentation method proposed by Wallner et al. [6], 12–24 min of Marker-based watershed transformation method proposed by Fan et al. [9], 4 min of CNN proposed by Ibragimov et al. [12] and 2.5 min of 2.5D network proposed by Qiu et al. [14], the proposed network significantly reduced segmentation time and improved segmentation efficiency.

In traditional automatic methods, the methods proposed by Gass et al. [7], Pei et al. [8] and Taghizadeh et al. [10] were all time-consuming in their iterative process, and their Dices of segmentation accuracy were not as high as our method. Although the Dice of segmentation accuracy reached $0.97 \pm 1\%$ in the method of Fan et al. [9], it was time-consuming. Compared with automatic segmentation networks, the method of Ibragimov et al. [12] didn't directly segment mandible automatically at the pixel level, which not only required a long time but also rough edges on their results. The methods proposed by Yan et al. [13] and Qiu et al. [14] were segmentation on the 2D images ignoring the

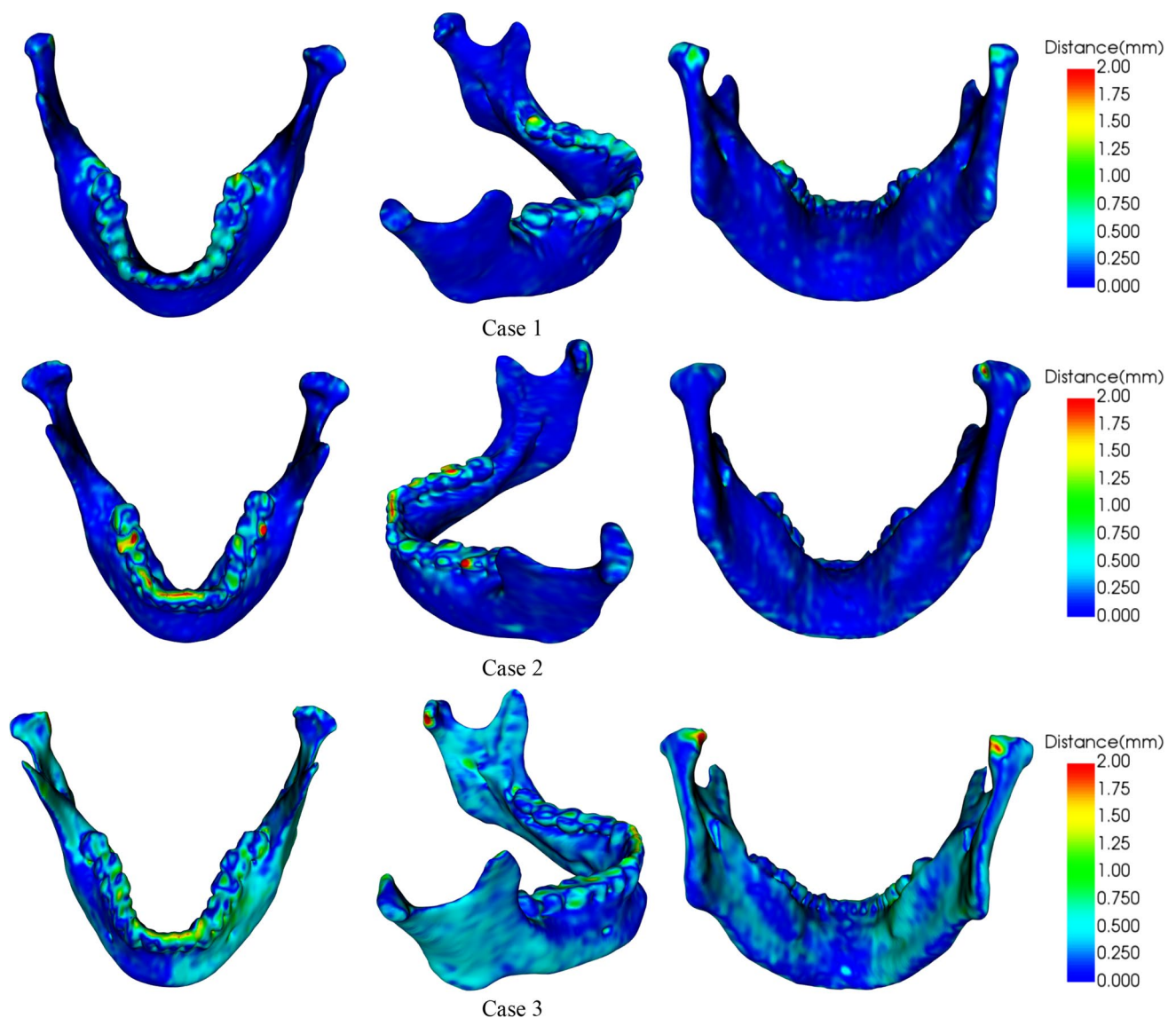


Fig. 5 presents the surface distance error map between ground truths and segmentation results of our method from three selected CT scans. For each case, three figures are showed from different perspectives (front, lateral, and back)

correlation between each CT slice. Compared with the Dice of 92.02% obtained by Yan et al. [13], 93.28% obtained by Qiu et al. [14], 94% obtained by Gou et al. [15], our method shows a better segmentation accuracy.

In summary, the proposed automatic segmentation network has accurately realized the automatic segmentation of mandible, and the segmentation accuracy is better than other methods. Although the proposed network has improved the segmentation performance of the mandible, there are still some drawbacks: firstly, there will still be some misjudgments at the blurred boundary of teeth; secondly, it still lacks some CT data from complex cases including pathological cases, patients with partial denture, edentulous patients with atrophied bone, and image artifacts caused by dental filling.

The proposed method can be used in some clinical applications. For example, in maxillofacial plastic surgery, it can be used to determine the mandibular angle and osteotomy trajectory in preoperative planning. In the implant placement simulation for the mandible, it can be used for statistical modeling and analysis [24]. In mandibular reconstruction, it can be also used to analyze the positions of some important anatomical landmarks in normal mandible for the improvement of the accuracy of surgery [25]. In the future, further research will be conducted by expanding relevant CT data and further optimizing the network.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent There was no informed consent required for the work reported in this manuscript.

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